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A coupled electrocatalytic system with reduced energy input for CO₂ reduction and biomass valorization†

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Coupling the cathodic electrochemical CO₂ reduction (eCO₂RR) with the anodic electrochemical 5-hydroxymethylfurfural oxidation reaction (HMFOR) is an appealing approach to co-produce value-added chemicals and simultaneously lower the energy input, while the main challenge is to develop effective electrocatalysts for eCO₂RR and HMFOR. Inspired by the techno-economic analysis, we herein report a coupling system for economical formate generation and biomass valorization by concurrent cathodic eCO₂RR and anodic HMFOR. Single Cu atom doped Bi (Cu₁Bi) and NiCo layer doubled hydroxides (NiCo LDHs) are used as the cathodic and anodic catalysts, respectively. Due to the accelerated water dissociation, the Cu₁Bi cathode catalyst demonstrated remarkably high activity ($j_{\text{formate}} > 1 \text{ A cm}^{-2}$) for formate conversion as well as robust stability (200 mA cm⁻² for 45 h). Meanwhile, the NiCo LDH anode displays high activity for HMF electrooxidation to 2,5-furandicarboxylic acid with a faradaic efficiency of over 95% at low potential. Consequently, the paired eCO₂RR-HMFOR system reduces the electricity input for the eCO₂RR-to-formate conversion to ~3493 kW h per tonne of formate, corresponding to 22.9% in energy saving compared to the conventional eCO₂RR (~4528 kW h per tonne of formate). This study offers an appealing route for CO₂ reduction and biomass valorization with low energy consumption.

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Broader context

Renewable energy-powered electrochemical CO₂ reduction (eCO₂RR) offers an appealing approach for both mitigation of global warming and the production of valuable chemicals. Conventional eCO₂RR is generally paired with the oxygen evolution reaction (OER). However, large overall energy consumption is required due to the sluggish reaction kinetics of the OER. Moreover, the anodic OER product, O₂, hardly contributes to economic feasibility from the viewpoint of its current value. In this study, we developed a coupling system consisting of the cathodic eCO₂RR and the anodic electrochemical 5-hydroxymethylfurfural oxidation reaction (HMFOR) for economical formate generation and biomass valorization. Single Cu atom doped Bi (Cu₁Bi) and NiCo layer doubled hydroxides were used as the cathode and anode catalysts, respectively. Cu₁Bi shows an industrially relevant formate production rate ($j_{\text{formate}} > 1 \text{ A cm}^{-2}$ at -0.9 V) and robust stability, outperforming most of the previously reported formate-producing catalysts. The coupled eCO₂RR-HMFOR system achieves a cathodic formate production (FE_{formate} ≈ 90%) and an anodic 2,5-furandicarboxylic acid production (FE_{FDA} ≈ 80%) at 237 mA cm⁻² with a cell voltage of 2.7 V, which is 0.8 V lower than the conventional eCO₂RR-OER system. This work opens a pathway for other prospective applications for the generation of commodity chemicals simultaneously on both electrodes in one electrolyzer.

Introduction

As renewable electrical power generation costs have continuously decreased and become cost-competitive with fossil fuels, the electrochemical CO₂ reduction reaction (eCO₂RR) into basic chemical feedstocks has been perceived as a promising method to store renewable electricity while abating carbon emissions.^{1–4} Among various products obtained from eCO₂RR, formate/formic acid may still be the near-term target product to achieve market/cost competitiveness because of its highest value per electron.⁵

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Formic acid is also widely explored as a hydrogen storage material in fuel cell applications and an essential building block in the leather tanning and textile industries with increasing market size (Fig. S1, ESI†).^{6,7} Among a variety of formate-selective electrocatalysts, Bi has stood out due to its relatively low cost and promising catalytic performances.^{8,9} Various strategies such as element doping, morphology control, and hybridizing with carbonaceous support have been developed to improve the performance of Bi-based catalysts.^{10–16} Despite a high FE of formate ($\text{FE}_{\text{formate}} > 90\%$) having been achieved, the partial current densities of formate (j_{formate}) for previously reported Bi-based catalysts were generally below 500 mA cm^{-2} , which is insufficient to achieve a desirable profit.

Recent evidence suggests that proton-feeding plays a vital role in eCO_2RR because the dissociation of H_2O to generate H^+ in a neutral/alkaline medium is rather sluggish.^{17–20} Therefore, a rational design of Bi-based catalysts that could facilitate water dissociation to provide sufficient H^+ for the protonation step would markedly boost the formate production rate. However, Bi itself is inert for water dissociation due to its poor adsorption of H^+ . Single-atom alloys (SAAs) with foreign atoms atomically dispersed in the matrix of a host metal can tune the electronic structure of the host metal and thus adjust the binding strength of reaction intermediates.^{21–23} We expected to introduce a secondary metal site into the Bi matrix to facilitate the water dissociation step.

On the other hand, the present cathodic eCO_2RR is coupled with the anodic oxygen evolution reaction (OER). However, the sluggish OER kinetics results in a large overall energy input and produces minimal valuable product (O_2), which does not contribute to economic feasibility.^{24–26} Replacing the OER with energy-favorable organic oxidation reactions (OORs) is an appealing strategy to address these challenges.^{27–30} This anodic substitution strategy can reduce the total input energy and maximize profitability by producing value-added by-products at the anode.^{31,32} Electrochemical oxidation of 5-hydroxymethylfurfural (HMFOR) is particularly appealing because the organic feeds can be derived from biomass waste.³³ Furthermore, the oxidation product 2,5-furandicarboxylic acid (FDCA) has a wide range of applications in green chemicals and biopolymers.^{34,35} It is a promising substitute for petroleum-based terephthalic acid in the production of polyethylene terephthalate (59%, Fig. S2, ESI†).³⁶ To evaluate the feasibility of the eCO_2RR -to-formate coupled HMFOR-to-FDCA system, we performed a preliminary techno-economic analysis (TEA) using a simplified model.³⁷ It is suggested that the conventional eCO_2RR -coupled OER (eCO_2RR -OER) system is economically infeasible due to the high electricity cost and low value of O_2 (Fig. S3a, ESI†). By contrast, the eCO_2RR -HMFOR system (Fig. S3b, ESI†) shows desirable economic profitability with net revenues of $\sim \$545$ for production per tonne of formic acid under commercially relevant performances (Table S1, ESI†).

To this end, we report a co-electrolysis system with cathodic eCO_2RR -to-formate on single Cu atom doped Bi (Cu_1Bi) coupled with anodic HMFOR-to-FDCA on NiCo layer doubled hydroxides grown on nickel foam (NiCoLDH/NF) to produce

value-added chemicals. The Cu_1Bi cathode catalyst demonstrated an industrially relevant formate production rate ($j_{\text{formate}} > 1 \text{ A cm}^{-2}$ at -0.9 V) while maintaining a high $\text{FE}_{\text{formate}}$ of 93% at 200 mA cm^{-2} over 45 h, outperforming most of the previously reported formate-producing catalysts. We discovered that the single Cu atom significantly promotes water dissociation thereby boosting formate production. The coupled eCO_2RR -HMFOR system achieves a cathodic formate production ($\text{FE}_{\text{formate}} \approx 90\%$) and an anodic FDCA production ($\text{FE}_{\text{FDCA}} \approx 80\%$) at 237 mA cm^{-2} with a cell voltage of 2.7 V , which is 0.8 V lower than the conventional eCO_2RR -OER system. This translates to about a 22.9% reduction in energy requirement.

Results and discussion

Synthesis of cathodic Cu_1Bi catalyst for eCO_2RR

The Cu_1Bi catalysts were synthesized through a two-step hydrolysis and *in situ* electrochemical reduction processes (Fig. 1a). Typically, Cu-doped layered basic bismuth nitrate- $\text{Bi}_6\text{O}_6(\text{OH})_3(\text{NO}_3)_3 \cdot 1.5\text{H}_2\text{O}$ (Cu-BiOON) was obtained by hydrolyzing commercial bismuth nitrate. The X-ray diffraction (XRD) pattern of Cu-BiOON was well assigned to the tetragonal basic bismuth nitrates without forming any Cu species (Fig. S4, ESI†). Scanning electron microscopy (SEM) and transmission electron microscopy (TEM) images showed that the Cu-BiOON exhibited hierarchical microflower morphology assembled *via* very thin nanosheets (Fig. S5a–c, ESI†). High-resolution TEM images (HRTEM, Fig. S5d and e, ESI†) revealed obvious lattice spacing of 0.37 nm , corresponding to the (101) planes of tetragonal BiOON. Scanning transmission electron microscopy–energy dispersive spectroscopy (STEM-EDS) elemental mapping exhibited the homogenous distributions of Bi, Cu, Br, O, and N (Fig. S5f and g, ESI†), indicating the successful incorporation of Cu. The Cu-BiOON was then electrochemically reduced into Cu_1Bi and the nanosheet structure was well maintained (Fig. 1b–d). All the diffraction peaks in the Cu_1Bi XRD pattern were assigned to metallic Bi (Fig. S4, ESI†), consistent with the X-ray photoelectron spectroscopy (XPS) results (Fig. S6, ESI†). The successful reduction of Bi was also confirmed by the operando X-ray absorption spectroscopy (XAS) where Bi L-edge overlapped with the metallic Bi reference under the applied potentials (Fig. S7, ESI†). The HRTEM image shown in Fig. S8b (ESI†) revealed an obvious interplanar spacing corresponding to the (012) plane of rhombohedral Bi. EDS exhibits a uniform spatial distribution of Bi and Cu (Fig. S8c, ESI†). The Cu content was determined to be 1.02 wt% by an inductively coupled plasma-optical emission spectrometer (ICP-OES, Table S2, ESI†).

The atomic structure of the Cu_1Bi catalyst was further investigated by high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) combined with EDS. Fig. 1e demonstrates the absence of Cu clusters or particles, indicating the possible atomic dispersion of Cu atoms in the Bi matrix. The isolated dark dots in Fig. 1f (yellow circle) can be ascribed to individual Cu atoms according to the Z-contrast difference between the light Cu and heavier Bi

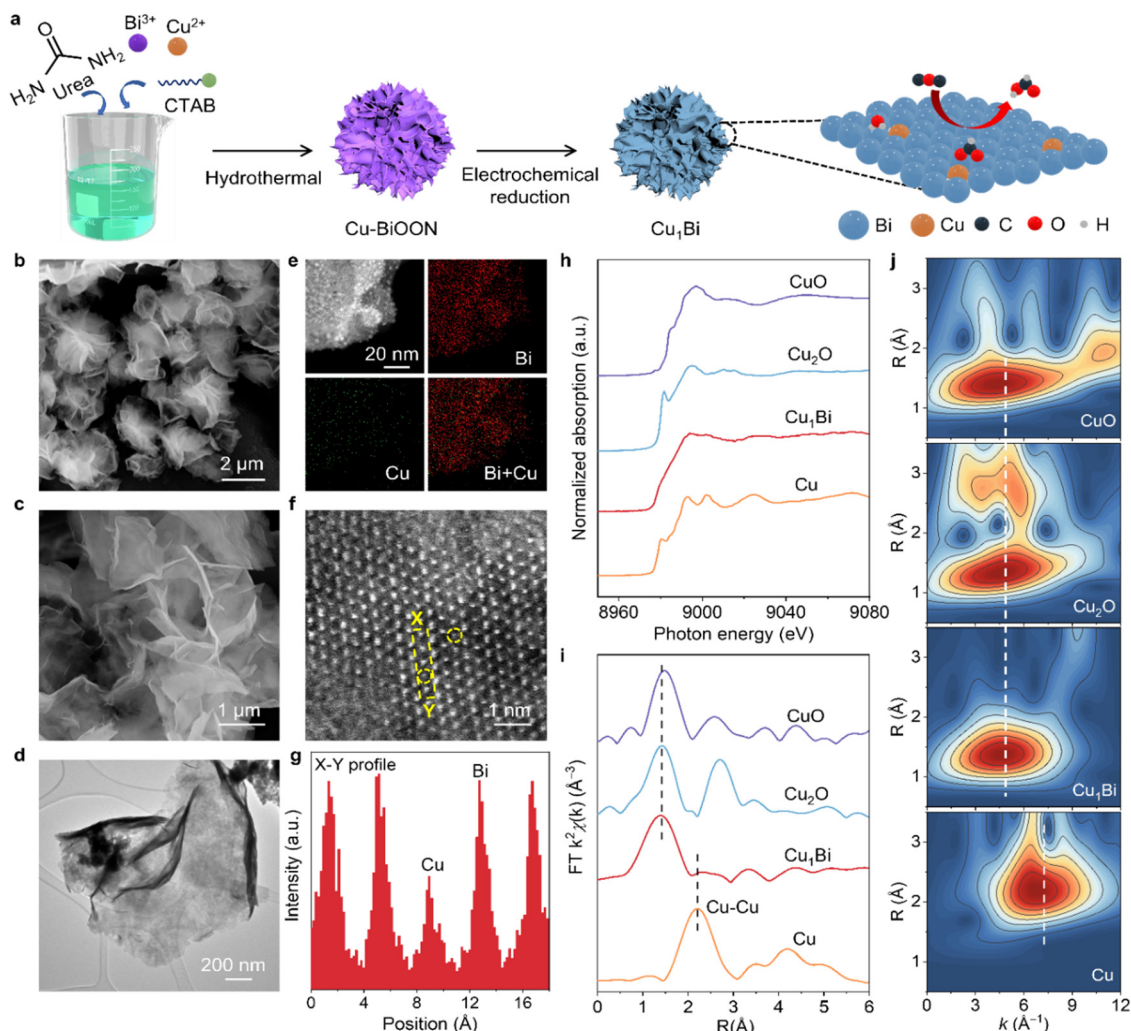


Fig. 1 (a) Illustration of the synthesis route of Cu_1Bi . (b), (c) SEM and (d) TEM and images of Cu_1Bi . (e) STEM-EDS mapping of Cu_1Bi . (f) HAADF-STEM image and (g) the corresponding intensity line profile for Cu_1Bi . (h) Normalized Cu K-edge XANES and (i) EXAFS spectra of Cu_1Bi and reference samples. (j) WT k^2 -weighted EXAFS contour plots of Cu_1Bi and reference samples.

atoms,³⁸ which is further corroborated by the corresponding line intensity profile (Fig. 1g), where the Cu site had a lower intensity than the Bi sites. We further resorted to X-ray absorption near-edge structure (XANES) and extended X-ray absorption fine structure (EXAFS) measurements to investigate the atomic structure of the Cu_1Bi catalyst. Cu K-edge EXAFS spectrum of Cu-BiOON shows a peak at about $1.47 \text{ \AA}$ (Fig. S9, ESI[†]) attributed to the Cu–O bond, suggesting that the Bi atom in Cu-BiOON is substituted by Cu. Cu K-edge XANES spectra show that the Cu K-edge absorption edge of Cu_1Bi is located between those of Cu and Cu_2O (Fig. 1h), implying that the surface-exposed Cu atoms are not over-oxidized. As shown in Fig. 1i, the EXAFS curve of the Cu_1Bi shows an obvious coordination shell at $1.49 \text{ \AA}$ attributed to the Cu–O bond.³⁹ The absence of Cu–Cu at $2.15 \text{ \AA}$ and Cu–O–Cu coordination shells at $2.65 \text{ \AA}$ precludes the existence of Cu and $\text{CuO/Cu}_2\text{O}$ particles. To gain more insights into the coordination environment of the metal atoms, the EXAFS plots were transformed using wavelet transform (WT), which can reflect R -space and k -space at the same time (Fig. 1j). A pronounced signal was

observed at $4.5 \text{ \AA}^{-1}$ in Cu_1Bi , assignable to the Cu–O coordination. Meanwhile, no intensity maximum at $7 \text{ \AA}^{-1}$ associated with the Cu–Cu was detected in Cu_1Bi , further verifying the atomically dispersed Cu atom. In addition, Cu K-edge EXAFS of Cu_1Bi under an applied potential (-0.8 V) demonstrated a prominent peak at $2.04 \text{ \AA}$ attributed to the Cu–Bi bond (Fig. S10, ESI[†]), suggesting that the catalysts were in the metallic state under an eCO_2RR environment. To this end, with all the above characterization results, it is clear that Cu_1Bi SAAs have been successfully synthesized.

eCO_2RR performances of Cu_1Bi

The eCO_2RR catalytic activity of the Cu_1Bi catalyst in a flow cell configuration (Fig. S11, ESI[†]) was examined using 1 M KOH as the electrolyte. Liquid and gaseous products were quantified by ionic chromatography and online gas chromatography, respectively. The linear sweep voltammetry (LSV) of Cu_1Bi in the CO_2 environment showed a much higher current density than that in the Ar environment, indicating the participation of CO_2 gas

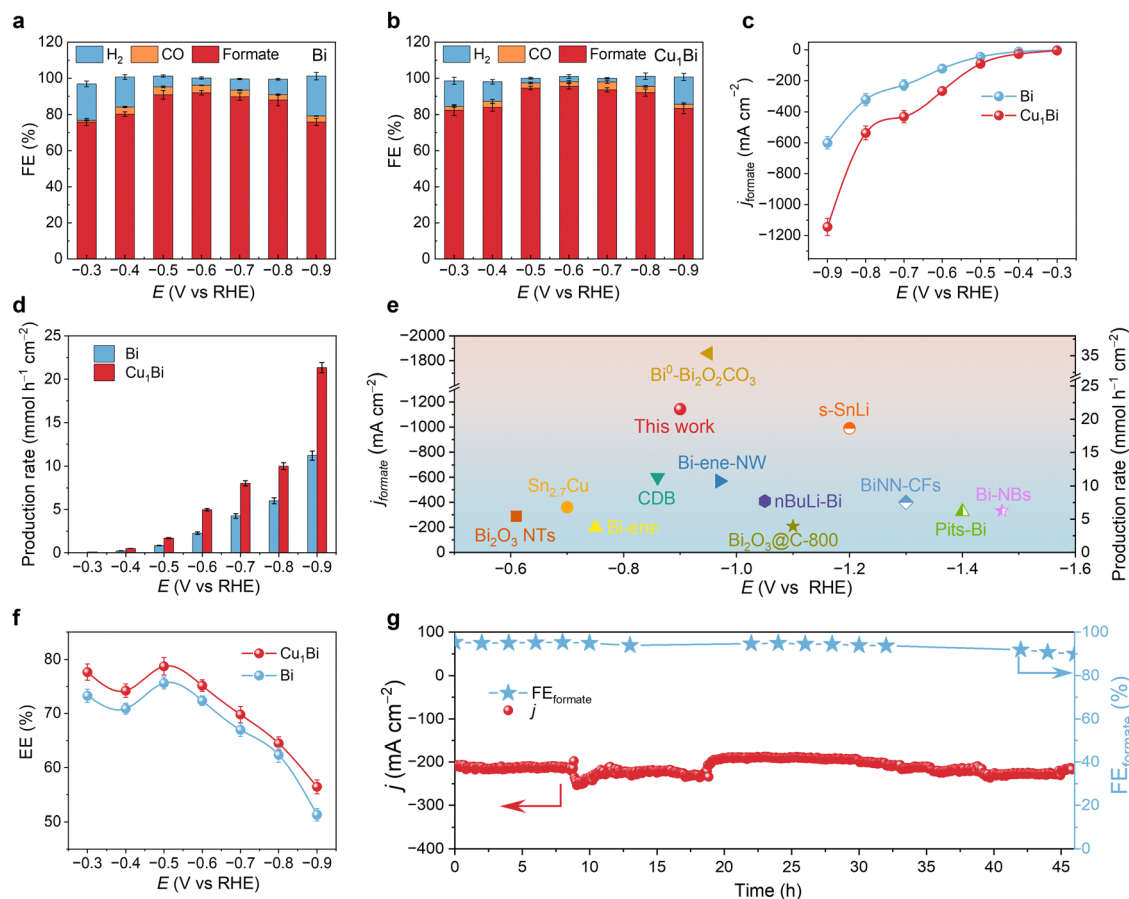


Fig. 2 FEs of all eCO₂RR products at different potentials of (a) Bi and (b) Cu₁Bi. (c) Formate partial current density against applied potential over Bi and Cu₁Bi. (d) Formate production rate of Bi and Cu₁Bi. (e) Comparison of the performance metrics of the Cu₁Bi with recently reported Bi-based electrocatalysts. (f) Energy efficiencies for eCO₂RR-to-formate conversion on Bi and Cu₁Bi. (g) Long-term operation stability of Cu₁Bi at -0.58 V.

in the reaction (Fig. S12, ESI†). At the applied potentials ranging from -0.3 to -0.9 V, it was found that formate was the major product for Cu₁Bi and Bi, together with small percentages of H₂ and CO (Fig. 2a and b). Significantly, Cu₁Bi showed a maximum FE_{formate} of up to 95.5% at -0.6 V, superior to that of Bi (91.8%). Meanwhile, a high plateau of FE_{formate} over 90% was retained in a wide potential window from -0.5 to -0.8 V. Simultaneously, the j_{formate} on the Cu₁Bi catalyst was dramatically increased compared with that on the Bi catalyst, achieving a higher j_{formate} of -536.2 mA cm⁻² with a high FE_{formate} of 92% (Fig. 2c). Notably, with the FE_{formate} still maintaining a high value of 83.5% at -0.9 V, the j_{formate} of Cu₁Bi further increased up to -1143 mA cm⁻², yielding a high formate production rate of 21.34 mmol h⁻¹ cm⁻² (Fig. 2d). This combination represents one of the highest values reported to date in a KOH environment (Fig. 2e and Table S3, ESI†). The formate cathodic energy efficiencies (EE) are displayed in Fig. 2f. The Cu₁Bi catalyst exhibits an EE of 75% at -266 mA cm⁻², outperforming most of the previously reported catalysts (Fig. S13, ESI†). The intrinsic activity of Bi and Cu₁Bi was compared by normalizing j_{formate} with respect to electrochemical surface area (ECSA). The ECSA-corrected j_{formate} of Cu₁Bi was still much higher than that of Bi, demonstrating that Cu₁Bi possesses a higher

intrinsic activity (Fig. S14, ESI†). In addition, according to the Nyquist plots (Fig. S15, ESI†), Cu₁Bi discloses a charge-transfer resistance (R_{ct}) of 1.8 Ω , lower than that of Bi (2.8 Ω), which suggests that Cu₁Bi embraces a faster charge-transfer rate. The long-term stability of the Cu₁Bi was evaluated at -0.58 V for 45 h, with an average j of 200 mA cm⁻² and FE_{formate} of around 92%, suggesting its remarkable stability for long-term operation (Fig. 2g). Post-catalysis morphology and structure characterizations revealed that the nanosheet morphology was well-retained and the Cu atoms were still atomically dispersed in the Bi matrix (Fig. S16 and S17, ESI†).

To highlight the important role of single Cu atoms for formate production over this Cu₁Bi catalyst, two control samples with varying Cu contents in the Cu₁Bi catalyst were prepared and their eCO₂RR performances were compared. The catalysts with high and low Cu contents were denoted as Cu₁Bi-H (2.82 wt% of Cu) and Cu₁Bi-L (0.64 wt% of Cu), respectively. SEM images indicated that both Cu₁Bi-L and Cu₁Bi-H possessed nanosheet morphology (Fig. S18a–d, ESI†), similar to that of the Cu₁Bi. The EXAFS curve suggested that the Cu atoms were still single-atom isolated in the Bi matrix of Cu₁Bi-L, while the Cu atoms in Cu₁Bi-H mainly existed in the aggregated form (Fig. S18e, ESI†). For Cu₁Bi-L, the formate was still the major product with a maximum

FE_{formate} of 94.6% at -0.6 V (Fig. S18f, ESI†). However, j_{formate} was lower than that of Cu_1Bi , likely due to the inadequate density of Cu single atoms. In sharp contrast, the $\text{Cu}_1\text{Bi-H}$ catalyst only achieved a maximum FE_{formate} of 88% at -0.5 V (Fig. S18g, ESI†). Further applying a more negative potential led to the drastic drop of FE_{formate} below 50% accompanied by significant hydrogen evolution. With this low selectivity, the j_{formate} of $\text{Cu}_1\text{Bi-H}$ decreased to only 410 mA cm^{-2} . The significant decrease in eCO_2RR performances over the $\text{Cu}_1\text{Bi-H}$ could be attributed to the aggregated Cu that favors the competitive HER.⁴⁰ In light of these control experiments, it can be inferred that the isolated Cu atoms play a key role in boosting the formate production rate of neighboring Bi sites. A certain amount of Cu atoms is required to sustain a high rate of formate production. If the density of Cu atoms is too low, a large portion of the Bi cannot be activated, resulting in low activity. If the density of Cu atoms is too high and Cu aggregation occurs, a competitive HER will dominate.

eCO_2RR performance enhancement mechanism

To reveal the role of Cu single atoms in promoting formate production, DFT calculations were performed on Bi and Cu_1Bi . The optimized adsorption configurations of intermediates are displayed in Fig. S19 (ESI†). We first investigated the eCO_2RR -to-formate pathway, and the corresponding free energy diagram is shown in Fig. 3a. For Cu_1Bi , the formation energy of $^*\text{OCHO}$ is 0.78 eV , which is much lower than that of Bi (1.67 eV). Therefore, the incorporation of single Cu atoms into Bi substantially facilitates eCO_2RR activity towards formate production. To reveal the important role of water dissociation for eCO_2RR , the adsorption configurations of intermediates for water dissociation are

presented in Fig. S20 (ESI†). Compared to the end-on adsorption on Bi, the H_2O^* and $^*\text{OH}$ are preferably adsorbed on the interstitial site between Bi and Cu with a shorter Cu–O bond. This suggests that the Cu site is indeed involved in the water dissociation process. As a key descriptor for evaluating the proton-feeding ability for formate production *via* eCO_2RR , the free-energy change (ΔG) of water dissociation is simulated. As shown in Fig. 3b, the ΔG over Cu_1Bi (0.62 eV) is significantly lowered as compared to that of Bi (1.77 eV), indicating that single Cu incorporation accelerated the water dissociation step. We then measured the kinetic isotope effect (KIE) of Bi and Cu_1Bi to evaluate the reaction kinetics of water dissociation and protonation process. As shown in Fig. 3c, when the solvent changes from H_2O to D_2O in 1 M KOH , the formation rate of formate decreases. The KIE of H/D is defined as the ratio of formate production rates in H_2O and D_2O . The KIE value of Bi was calculated as 1.8, which is a characteristic value of primary KIE.⁴¹ This indicates that the sluggish water dissociation over Bi limits its eCO_2RR kinetics. By contrast, the KIE over Cu_1Bi decreases close to 1, which corroborates the promoted protonation process through accelerating water dissociation and agrees well with the above DFT results. To further determine the role of Cu, we simulated the density of states (DOS) of Bi and Cu_1Bi (Fig. S21, ESI†). The Cu_1Bi shows a higher occupied state than the bare Bi near the Fermi level, indicating stronger bonding between catalyst and adsorbate. It is reported that the water dissociation ability is correlated with $^*\text{OH}$ adsorption, which can be evaluated by carrying out LSV in Ar-saturated 0.1 M KOH solutions.⁴² Clearly, Cu_1Bi exhibits a more positive peak compared with Bi (Fig. 3d), which implies that the single Cu

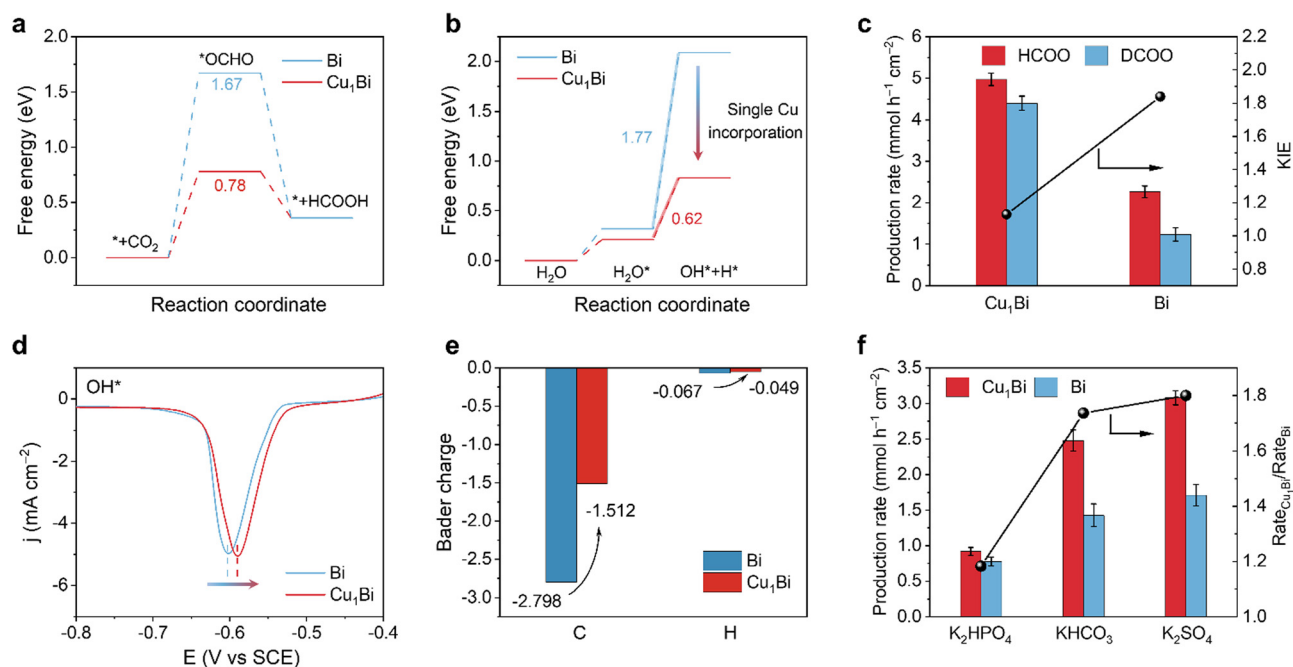


Fig. 3 (a) Gibbs free energy diagrams for HCOOH production on Bi and Cu_1Bi . (b) Gibbs free energy diagrams for the H_2O dissociation process on Bi and Cu_1Bi surfaces. (c) KIE of H/D over Bi and Cu_1Bi catalysts at -0.6 V RHE . (d) LSV curves of Bi and Cu_1Bi in Ar-saturated 0.1 M KOH . (e) Bader charge of C and H in $^*\text{OCHO}$ on Bi and Cu_1Bi models. (f) Formation rates of formate and the ratio of formate formation rate over Bi and Cu_1Bi in different electrolytes.

introduction strengthens the OH* adsorption, thereby promoting water dissociation. Bader charge analysis of *OCHO was further performed on Bi and Cu₁Bi models (Fig. 3e). The Bader charges assigned to C^{δ+} and H^{δ+} on Cu₁Bi were −1.512 and −0.049, respectively. Their absolute value was lower than both those on Bi (−2.798 and −0.067), suggesting the loss of fewer electrons. These changes resulted in the electron localization around C^{δ+} within *OCHO, enabling its interaction with the nearby adsorbed *H to generate formate.

To further highlight the important role of water activation in eCO₂RR, the pH effect of electrolytes was investigated (Fig. 3f). It is well known that the local pH at the electrode/electrolyte interface increases in the order of K₂HPO₄ < KHCO₃ < K₂SO₄⁴³ and thus we adjusted the local pH with these three different electrolytes. An increasing trend of formate production rate was observed over Bi and Cu₁Bi when the electrolyte changed from K₂HPO₄ to KHCO₃ and then to K₂SO₄. This indicates that higher local pH favors the production rate of formate, which is consistent with previous work.³⁸ Furthermore, the ratio of formate production rates between the Cu₁Bi and Bi

catalysts (Rate_{Cu₁Bi}/Rate_{Bi}) was calculated, and it is shown that its enhancement over Cu₁Bi was more significant than that over Bi, suggesting that the role of single Cu atoms in enhancing the formate production is more significant at a higher pH value. This further supports our speculation that the single Cu atoms enhance formate formation by accelerating the water activation, which becomes more difficult at a higher pH.

Synthesis of anode NiCoLDH/NF catalyst for the HMFOR

In parallel with the Cu₁Bi catalysts for cathodic eCO₂RR-to-formate conversion with ultra-high formate production rate, NiCo layer doubled hydroxides *in situ* grown on nickel foam (NiCoLDH/NF) have been adopted as efficient anodic catalysts for HMFOR to match the cathodic reaction. The NiCoLDH/NF was synthesized firstly by growing NiCo metal-organic framework on NF (NiCoMOF/NF) *via* a hydrothermal method, followed by a cyclic voltammetry-assisted hydrolysis approach to form NiCoLDH. SEM image showed that the NiCoMOF precursor presented a stacked nanosheet morphology assembling on NF (Fig. S22, ESI[†]). After the hydrolysis process, the bulk NiCoMOF

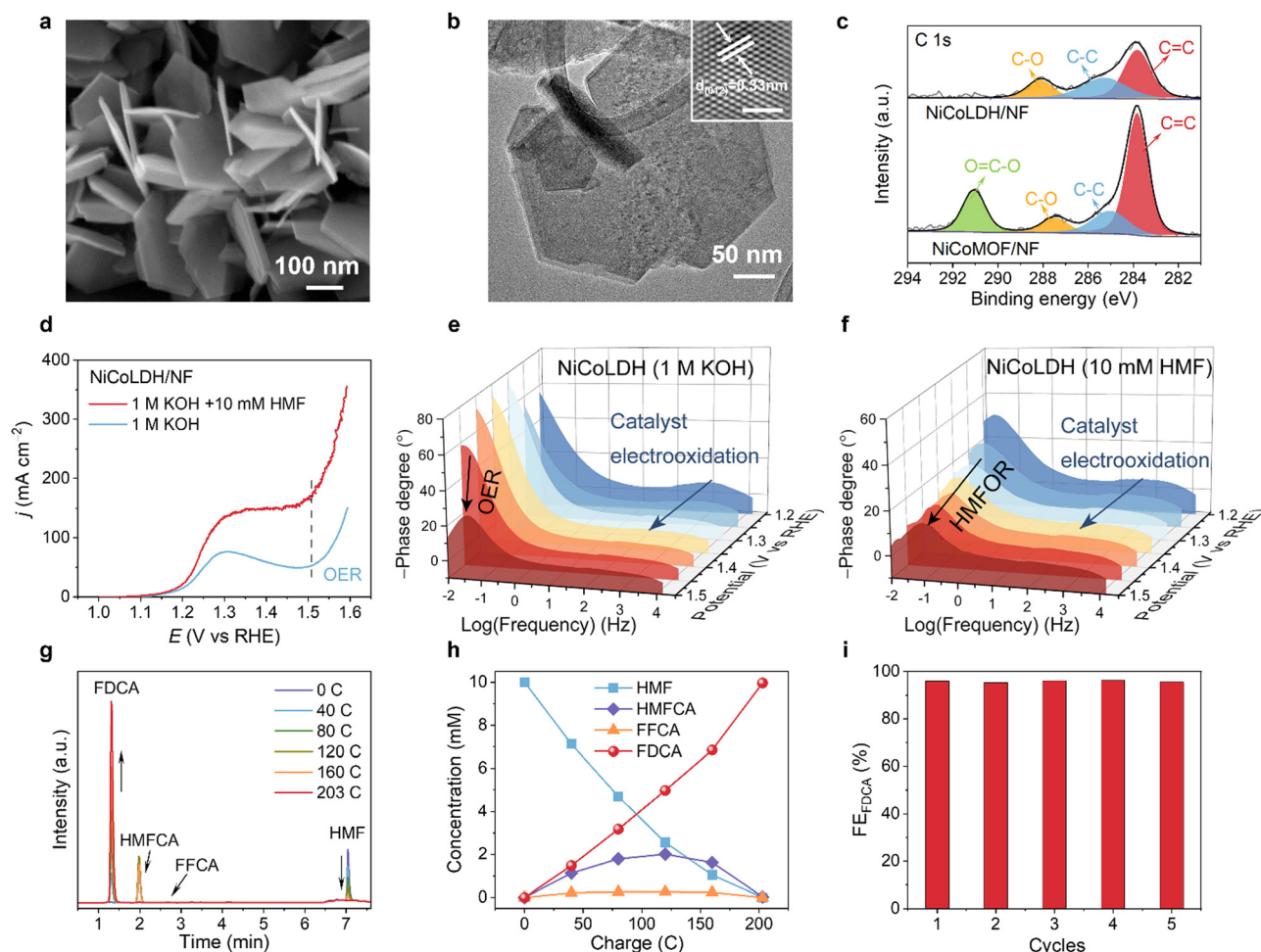


Fig. 4 (a) SEM images of the NiCoLDH. (b) TEM and HRTEM (insert) of NiCoLDH. Scale bar of inset: 1 nm. (c) High-resolution C 1s XPS spectra of NiCoMOF/NF and NiCoLDH/NF. (d) LSV curves of NiCoLDH/NF in 1 M KOH with and without 10 mM HMF. (e) and (f) Bode phase plots of NiCoLDH/NF in the electrolytes without and with 10 mM HMF. (g) HPLC-trace acquired at various charges. (h) Conversion and formation of oxidation products over passed charges. (i) FE_{FDCA} for five successive cycles with NiCoLDH/NF.

transformed into large amounts of nanoarrays consisting of hexagonal nanoplates with sizes between 100–200 nm (Fig. 4a and Fig. S23, ESI†). The nanoplate morphology of NiCoLDH was further verified by TEM. The lattice spacing matched the (012) facet of NiCoLDH (Fig. 4b). Meantime, the EDS elemental mapping revealed that the Ni, Co, O, and C were uniformly distributed throughout the nanoplates (Fig. S24, ESI†). Such transformation was further evidenced by the XRD analyses. As shown in Fig. S25 (ESI†), the initial diffraction peaks of NiCoMOF completely disappeared after hydrolysis, and multiple new peaks emerged assignable to $\text{Co}(\text{OH})_2$ (PDF#30-0443) and $\text{Ni}(\text{OH})_2$ (PDF#14-0117). To further gain insight into the evolution from NiCoMOF precursors to NiCoLDH, XPS was utilized to examine the change of valence states. As compared to the C 1s of NiCoMOF (Fig. 4c), the peak attributed to the carboxylate ($\text{O}=\text{C}-\text{O}$) group completely disappeared over NiCoLDH, indicating the dissociation of terephthalic acid linker from the NiCoMOF surface.⁴⁴ The O 1s of NiCoMOF (Fig. S26a, ESI†) can be deconvoluted into three surface components resulting from M–O, $\text{O}=\text{C}-\text{O}$, and –OH of the terephthalic acid ligand, separately. In sharp contrast, only two types of peaks assigned to M–O and M–OH were observed in the O 1s spectra of NiCoLDH. This confirms the formation of metal hydroxides. The appearances of Ni^{3+} and Co^{3+} species in Ni 2p and Co 2p of NiCoLDH/NF further verified this transformation (Fig. S26b and c, ESI†).

The electrocatalytic performance of NiCoLDH/NF toward anodic HMFOR was investigated in a three-electrode setup. The catalytic activities of NiLDH/NF and CoLDH/NF for HMFOR were also evaluated for comparison. It is evident that NiCoLDH/NF shows the lowest onset potential and the most significant current response compared with NiLDH and CoLDH (Fig. S27, ESI†), confirming the facilitated HMFOR over NiCoLDH. Fig. 4d displayed the LSV curves of NiCoLDH/NF in 1 M KOH with and without 10 mM HMF. In the pure 1 M KOH solution, the NiCoLDH/NF showed moderate OER activity with a current density of 100 mA cm^{-2} at 1.57 V. While in 1 M KOH containing 10 mM HMF, the current density increased significantly and achieved 280 mA cm^{-2} at the same potential of 1.57 V. The Nyquist plots for NiCoLDH/NF at 1.35 V in 1 M KOH without 10 mM HMF showed nearly vertical lines, indicating a high charge transfer resistance for the OER (Fig. S28, ESI†). By contrast, the Nyquist plots of NiCoLDH in 10 mM HMF displayed a semicircle shape, suggesting that the existence of HMF accelerates charge transfer at the electrode–electrolyte interface. The Tafel plots were plotted to evaluate the kinetics of the anodic reaction catalyzed by NiCoLDH/NF (Fig. S29, ESI†). The Tafel slope of NiCoLDH/NF for HMFOR (80 mV dec^{-1}) was much smaller than that for the OER (156 mV dec^{-1}), suggesting a more favorable catalytic kinetics of the HMFOR. To further verify the difference in reaction kinetics for the HMFOR and OER, *in situ* electrochemical impedance spectra (EIS) tests were conducted for NiCoLDH/NF over a potential gradient from 1.2 to 1.5 V and the corresponding Bode phase plots are shown in Fig. 4e and f. For NiCoLDH under OER conditions, the peak above 10^1 Hz can be attributed to the catalyst electrooxidation⁴⁵ and the characteristic peak for OER starts to appear at 1.5 V, suggesting the sluggish kinetics of the OER.⁴⁶ By contrast, the phase angle between 10^{-1}

and 10^0 Hz starts to decrease at 1.25 V when HMF is added into the electrolyte, demonstrating the preference of the HMFOR over the OER, which is consistent with the LSV curves. Product selectivity is another important criterion for judging the feasibility of biomass upgradation since the high selectivity of value-added products can lower the separation cost. High-performance liquid chromatography (HPLC) was used to quantify the oxidized products of the HMFOR. The concentration of specific species was determined according to the calibration curves for HMF, HMFCa, FFCA, and FDCA (Fig. S30, ESI†). NiCoLDH/NF-catalyzed HMFOR was conducted in 35 mL 1 M KOH with 10 mM HMF at 1.32 V. Nearly 203 C of the theoretical passed charge is required to convert all the HMF to FDCA. The development of HPLC traces and the HMF/product concentration with the increasing quantity of charge (every 40 C) are presented in Fig. 4g and Fig. 4h, respectively. Clearly, with the quantity of passed charges increased gradually to 203 C, the concentration of HMF decreased to 0 while that of FDCA increased to maximum. Remarkably, the calculated FE of FDCA (FE_{FDCA}) at the end of electrolysis is as high as 96%, suggesting that NiCoLDH/NF is highly active and selective for HMF upgradation. The stability of NiCoLDH/NF for the HMFOR was evaluated by conducting five repetitive runs in 1 M KOH + 10 mM HMF solution at 1.32 V (Fig. S31, ESI†). As shown in Fig. 4i, the FE_{FDCA} is retained in all cycles and the nanoplate morphology is well maintained after the stability measurement (Fig. S32, ESI†). Therefore, the prepared NiCoLDH/NF catalysts are highly active and selective for HMFOR and could be able to couple with Cu_1Bi for the $\text{eCO}_2\text{RR-HMFOR}$ overall system.

Electrocatalytic performance of the coupled $\text{eCO}_2\text{RR-HMFOR}$ system

Encouraged by the excellent performance of both cathodic eCO_2RR -to-formate conversion over Cu_1Bi and anodic HMFOR-to-FDCA over NiCoLDH/NF, a two-electrode electrolyzer was constructed. Membrane-electrode-assembly (MEA) flow reactor with zero-gap configuration is known for minimizing the Ohmic resistance.^{47,48} Therefore, a MEA flow reactor was constructed for converting CO_2 into formate paired with biomass upgrading using Cu_1Bi and NiCoLDH/NF as cathodic and anodic catalysts, respectively (Fig. 5a). The LSV polarization curves for traditional CO_2 electrolysis ($\text{eCO}_2\text{RR-OER}$) and $\text{eCO}_2\text{RR-HMFOR}$ were acquired on this setup and are shown in Fig. 5b. With the addition of HMF to the anode feed, the onset potential was clearly reduced and the current densities for the $\text{eCO}_2\text{RR-HMFOR}$ system were significantly higher than that for the $\text{eCO}_2\text{RR-OER}$ system at any given voltage. Product analysis from the cathode for the two systems shows that the $\text{FE}_{\text{formate}}$ at different voltages is almost the same and can maintain above 90%, which indicates that anolyte does not affect the selectivity of the cathodic eCO_2RR . For the $\text{eCO}_2\text{RR-OER}$ system, cell voltages of 3.5 and 3.75 V were required to reach current densities of 237 and 280 mA cm^{-2} (Fig. 5c). By contrast, the required voltages to reach the same current densities decreased by 0.8 and 0.75 V to 2.7 and 3 V in the $\text{eCO}_2\text{RR-HMFOR}$ system (Fig. 5d). Thus, the paired $\text{eCO}_2\text{RR-HMFOR}$ system is able to reduce the electricity input for the eCO_2RR -to-formate

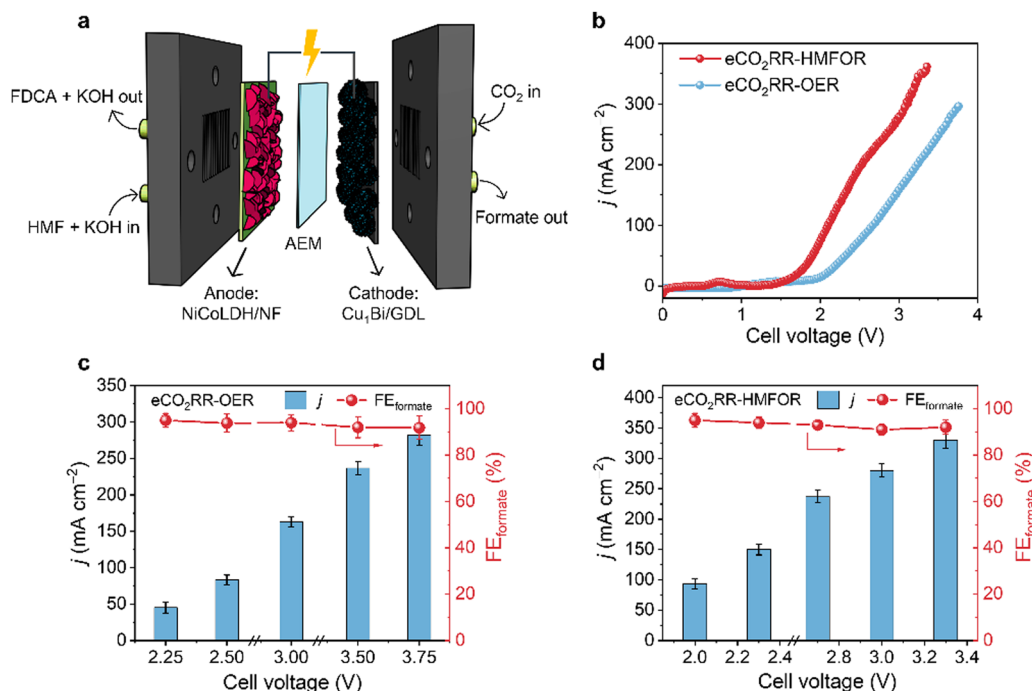


Fig. 5 (a) Schematic of the coupled eCO₂RR-HMFOR system in the MEA setup. (b) LSV curves for eCO₂RR-HMFOR and eCO₂RR-OER systems. (c) Current density and FE_{formate} as a function of cell voltage for the eCO₂RR-OER system. (d) Current density and FE_{formate} as a function of cell voltage for the eCO₂RR-HMFOR system.

conversion to ~ 3493 kW h per tonne of formate, corresponding to 22.9% energy saving compared to the conventional eCO₂RR-OER (~ 4528 kW h per tonne of formate). The reduced energy input can directly lower the electricity cost so that it is below the formate market price ($\sim \$349$ per tonne of formate for eCO₂RR-HMFOR *versus* $\sim \$452$ per tonne of formate for eCO₂RR-OER), thus achieving profitable formate production. Furthermore, the high-valued anodic FDCA products increase the overall profitability of the electrolyzer as a system. To evaluate the possibility of practical application, the electrochemical stability of the eCO₂RR-HMFOR system was investigated. The anolyte was refreshed every 2 h to prevent the degradation or Cannizzaro reaction of HMF. As shown in Fig. S33 (ESI[†]), the eCO₂RR-HMFOR system exhibits good stability with a high current density of 150 mA cm^{-2} for 10 h. Meanwhile, the average FE_{formate} and FE_{FDCA} can be maintained over 90% and 85%, respectively, demonstrating good stability. It is noteworthy that the downstream product separation, which is not covered in this work, could bring high costs to the system and harm the economic feasibility of large-scale applications.⁴⁹ More efforts will be devoted to other components of coupling system design, including cathode gas diffusion electrodes, anode catalysts, membranes, solid electrolytes, interfaces, and so on, to address the product separation and purification problem.

Conclusions

In this work, we have demonstrated a coupled eCO₂RR-HMFOR electrocatalytic system for formate production and biomass-derived HMF valorization with lower energy consumption.

Preliminary TEA highlights the economic feasibility of this coupled system with commercially relevant performances. A SAA electrocatalyst, Cu₁Bi, was developed for highly active cathodic eCO₂RR towards formate at $j_{\text{formate}} > 1 \text{ A cm}^{-2}$ with FE_{formate} $> 85\%$ in a flow cell. The advantageous performance for formate production can be attributed to the accelerated water dissociation process by the single Cu atoms, which facilitates the protonation step. The coupled electrocatalytic system is capable of obtaining valuable products of FDCA at the anode and formate at the cathode with a 22.9% reduction in electricity consumption as compared with the conventional eCO₂RR-OER system. This work not only offers an effective strategy for designing highly active electrocatalysts for eCO₂RR-to-formate conversion but also provides a promising avenue for profitable and sustainable biomass valorization.

Author contributions

S.-Q. Liu: conceptualization, methodology, data curation, formal analysis, investigation, visualization, and writing – original draft. M.-R. Gao: calculation. Y. Wang, L. Cui and S. Wu: data curation. R.-F. Feng: data curation. Y. Guo and X.-Z. Fu: writing – review and editing. J.-L. Luo: funding acquisition, resources, writing – review and editing and supervision.

Conflicts of interest

There are no conflicts to declare.

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